

RF Sim for KiCad

User Manual

RF matching networks and printed antennas,
simulated from the KiCad board with openEMS

Version 0.27.2 · November 2026

KiCad 10 · Windows 10 / 11 · openEMS 0.37

The screenshots show board #9 of a public antenna reference kit (dual-band PCB antenna, 868 / 915 MHz and 2.4 GHz) with its RF chain J9 – Z35 – Z38. A real design is walked through step by step in the separate *Worked Example*.

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1 What RF Sim does

RF Sim is a KiCad plugin. It builds a 3D electromagnetic model from the open board and solves it with openEMS (finite-difference time domain). It answers two questions:

- **RF chain:** which values do the matching parts need? The layout between the RF chip and the antenna is simulated once. You then move a slider per part and see S11, the Smith chart and the power to the antenna at once. **Autotune** searches E24 values or real vendor parts. **Apply to board** writes the values to the footprints.
- **Antenna:** how does the printed antenna perform on this board? Feed impedance, S11, radiation efficiency, directivity, realised gain and the 3D radiation pattern, optionally inside a plastic housing.

The simulated antenna impedance is the load of the RF chain. So the matching values are tuned for the real antenna on the real board, not for 50 Ω.

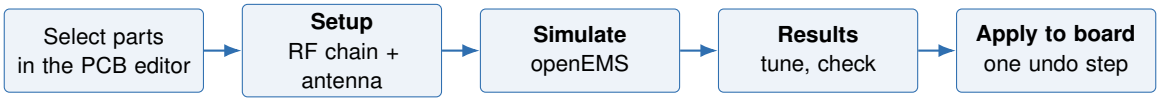


Figure 1: Workflow.

2 Installation

The separate *Installation Guide* describes the download, the plugin install, the KiCad settings, the first start, openEMS and a first test step by step. In short:

1. **Plugin. Plugin and Content Manager** → **Install from File…** → `kicad-rfsim-pcm-<version>.zip` → **Apply Pending Changes**.
2. **KiCad API. Preferences** → **Plugins**: enable the KiCad API, interpreter is KiCad's own `pythonw.exe`. Restart KiCad. The first start installs the plugin's Python packages (internet, a few minutes).
3. **openEMS.** On the first simulation RF Sim offers to download openEMS 0.37 (about 65 MB) into `%LOCALAPPDATA%\kicad-rfsim\openEMS`, or select an existing `openEMS.exe` on **Solver Settings**.

The **RF Sim** button then appears in the toolbar of the PCB editor. It opens one window per board. A second KiCad instance with another board gets its own window, so you can look at results while another board simulates.

> Updates: install the new zip the same way. Results of older versions stay usable. A result is computed again only when the layout, the setup or the solver settings changed.

3 Quick start

The separate *Worked Example* walks through a real design step by step, with a screenshot of every step. In short:

1. Open the board in the PCB editor. Press **RF Sim**.
2. In the PCB editor select the matching parts and one segment of the antenna trace after the last part.
3. **Setup**: **Read Selection**. Choose the **Band**. Check the antenna mode. Press **Create Setup from Selection**.
4. **Solver Settings**: *Fast* for a first look, *Automatic* or *Accurate* for final numbers.
5. Optional: **Inspect Mesh** shows the model as the solver will get it (a few seconds).
6. **Simulate**: tick the setups, **Start Simulation**. Plan minutes on *Fast* and hours on *Automatic*.
7. **Results**: the antenna report, then the RF chain. Move the sliders, or press **Autotune**.
8. **Apply to board**, confirm. Then **Tools** → **Update Schematic from PCB** in KiCad.

4 The RF Sim window

This chapter is the reference for each page. The *Worked Example* shows them in use. The window has five pages: **Setup**, **Casing & 3D**, **Simulate**, **Results** and **Solver Settings**. The status line at the bottom shows what runs and whether openEMS is ready.

4.1 Setup

Selection in the PCB editor. Select the parts you want to tune and press **Read Selection**. Only the selected parts get ports, and the model region shrinks to them. RF Sim then finds the ports itself:

- the unselected neighbour on the chip side becomes the **input port** (port 1);
- the antenna side becomes the **output port** (port 2, loaded with the antenna);
- other unselected neighbours (a test connector, a second branch) stay inside as open ports;
- unselected shunt parts to ground are fixed lumped parts.

Why explains the choice. **Input port / Output port** and **Apply Ports** change it without a new simulation. **Show in KiCad** selects the shown parts in the PCB editor.

Antenna. Four modes:

Detect automatically	finds the antenna copper on the RF nets and the feed pad; shows what was found and why.
Define antenna	you pick the antenna net, the feed pad and how the feed connects to ground (*ground plane below the pad* or *ground beside the pad*).
No antenna	RF chain only.
Antenna board	a soldered-in antenna PCB with its own KiCad project (e.g. a vertical antenna whose STEP is a footprint's 3D model). Placement check on **Casing & 3D**.

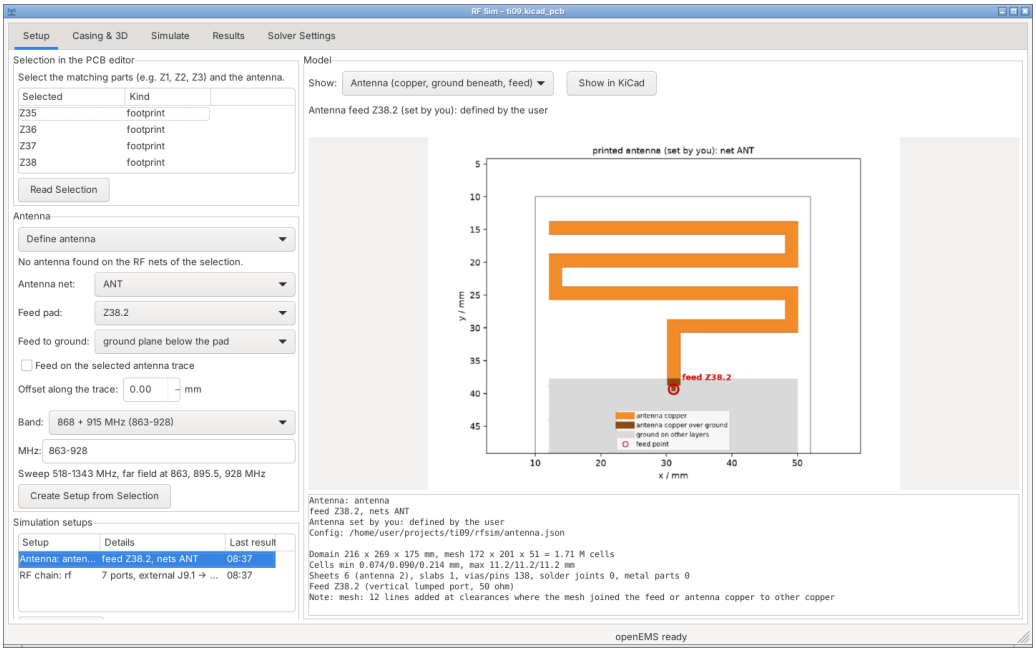


Figure 2: The antenna picture: antenna copper (orange), copper over ground (brown), ground on other layers (grey), feed point (red).

Feed on the selected antenna trace. With a pi network the last matching pad is a part pad. To feed the antenna where it starts, select the matching parts *and one segment of the antenna trace* after the last pad, then tick this box. The trace is cut there and a short port bridges the gap.

Band. The operating band of the design: a preset or MHz ranges such as `2400-2483.5` or `863-870, 2400-2483.5`. It sets the sweep, the far-field frequencies and the markers of all setups of the board (section 6). **Create Setup** needs a band.

Simulation setups lists the setups of the board with the last result. A click shows the model description below the picture. **Inspect Mesh** builds the selected setup's mesh (section 5.3).

The setups are stored in `<project>/rfsim/rf.json` and `antenna.json`.

4.2 Casing & 3D

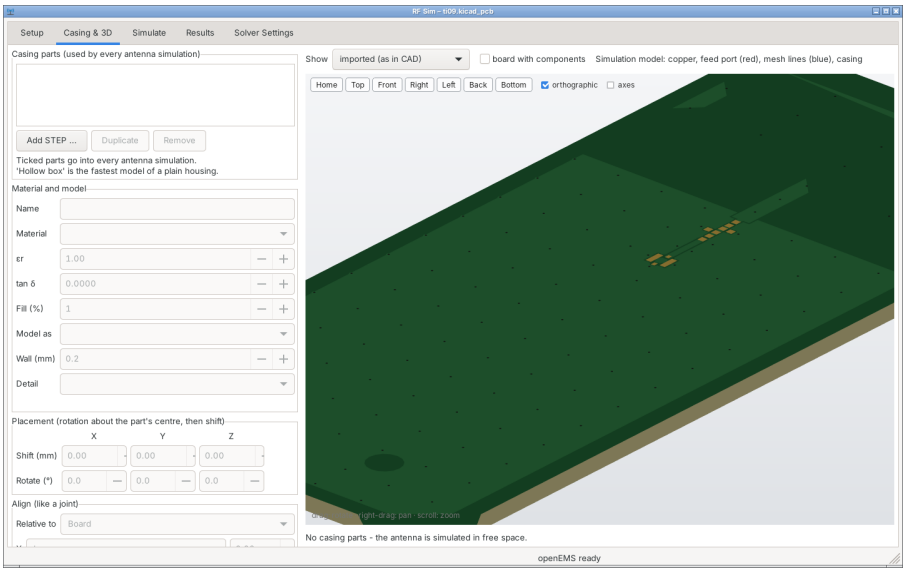


Figure 3: Casing & 3D: the board as KiCad exports it; housing parts are added here.

Put the board into its housing for the antenna simulation.

- **Add STEP . . .**: one or more STEP or STL files (e.g. shell and lid).
- **Material**: presets with ε_r and $\tan\delta$ (ABS, PC, PA66-GF30, PBT, POM, glass, metal, . . .). **Fill** below 100 % models infill or ribs.
- **Model as**: the tessellated *shape*, or a *hollow box* (outer size and wall) – the fastest model of a plain housing.
- **Placement** (shift, 90° rotation) and **Align** (centre on, align faces, stack on / under).
- **Show**: *imported* (as in CAD) or *simulation model* (what the solver sees).

Ticked parts take part in every antenna simulation. They are stored in `rfsim/enclosure.json`.

4.3 Simulate

- Tick the setups and press **Start Simulation**. RF Sim simulates a snapshot of the open board, including unsaved edits.
- An unchanged setup is not simulated again. **Run again even if layout and settings are unchanged** forces it.
- **Wave view: record fields** adds one run per setup for the animated field (section 5.4).
- **Whole-board wave view** joins the RF chain and the antenna into one animation (no extra run).
- **Run**: *Simulation*, or *Mesh convergence check* (section 5.6).
- **Progress**: run k of n, remaining time and the energy decay towards the stop criterion. **Stop** ends all solver processes at once.

Typical run time on a 4-core desktop	Fast	Automatic	Accurate
RF chain, 5 ports, small region	2–3 min	7–10 min	30–50 min
2.4 GHz printed antenna	1–5 min	5–15 min	15–75 min
Sub-GHz printed antenna	15–45 min	1–3 h	4–12 h
Vertical antenna board in a casing	20–45 min	2–3.5 h	8–12 h

The RF chain needs one run per port: one for each external port and one for each tuned part (a port across its two pads). Its time grows with the number of tuned parts and the region.

Simulation times are long. openEMS runs on the CPU only, with no GPU support. Plan minutes for a first look and hours for final numbers on an older PC. The estimate in the list is cautious: the real run usually ends sooner, because a run stops as soon as the fields have died down. Shorter run times are the main work in progress.

4.4 Results

The list on the left shows each result with its time and status. A failed plausibility check is marked. **Show** selects the page: *Result*, *How this was solved*, *Wave view*, *Mesh* or *Mesh convergence*.

Open in Browser	the same page in Chrome or the default browser.
Open Result Folder	`<project>/rfsim/results/<setup>/`.
Apply Values from File…	applies values exported from a tuning viewer opened in a browser.
Export S-Parameters…	Touchstone: antenna S11 (.s1p) or the RF chain's EM N-port.
Save Page as Single File…	the page with everything inside, to send or archive.
Delete Field Recordings	frees the space of the wave-view recordings.

4.5 Solver Settings

- **Band** and **Apply Band to All Setups**: change the operating band later.
- **Settings** per setup: *Automatic* (recommended), *Fast* (first look), *Accurate* (final numbers) or *Custom* (every value editable). Frequencies can be edited in every mode. Editing the sweep of one setup sets it in the other setups of the board.
- Below the grid: the meaning, default and setup value of the selected row.
- **openEMS**: path and version check.
- **Vendor part libraries**: a folder with measured capacitor and inductor S-parameters. The tuning viewer can then step through real parts.
- **Vendor S-parameter files for single parts**: a Touchstone file for an RF switch, balun, filter or chip antenna, assigned pad by pad.

5 Result pages

5.1 Tuning viewer (RF chain)

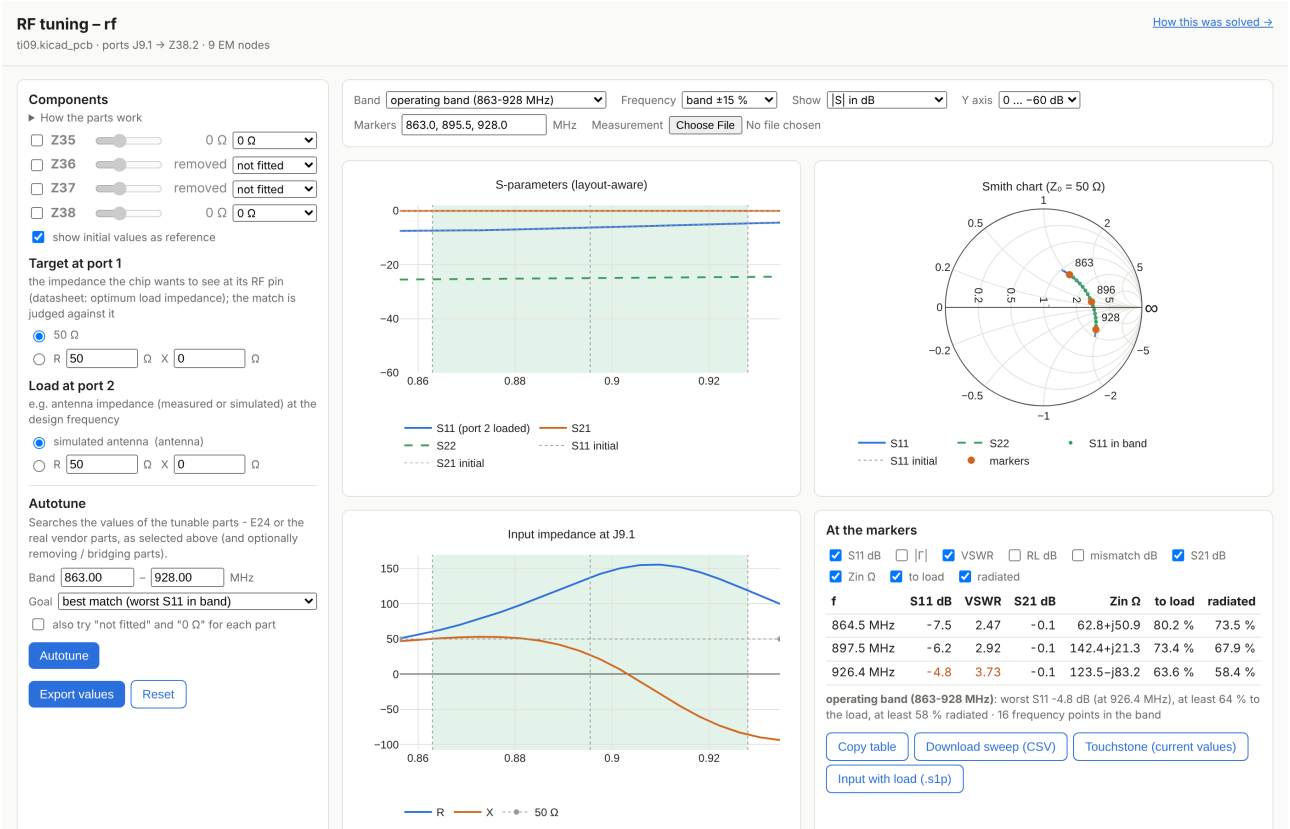


Figure 4: Tuning viewer: parts, S-parameters, Smith chart, input impedance and the marker table. The operating band is shaded.

- **Components**: a slider and a list per part. A C or L place can be *fitted as C*, *fitted as L*, *not fitted* or *0 Ω*. **Autotune** keeps each part's kind as set, or chooses C or L per part (**C / L**: *autotune chooses C or L per part*). **Part models**: ideal L/C with Q and ESL, or real vendor parts.
- **Band**: the board's operating band is preselected. It is shaded in the plots and sets the markers and the autotune band. Many other presets (LTE, LoRaWAN, GNSS, Wi-Fi, …) are available.
- **Show**: $|S|$ in dB or linear, VSWR, return loss or mismatch loss. **Y axis**: range.
- **Target at port 1**: the load impedance the RF chip wants (datasheet "optimum load"). S11, the table and autotune are

then judged against it.

- **Load at port 2**: the simulated antenna (default) or a fixed $R + jX$. A measured or vendor antenna file (.s1p) is set on **Solver Settings**.
- **Autotune**: searches the values for the best worst-case S11 in the band, or the most power to the antenna. Tick the box in front of a part to keep it fixed.
- **At the markers**: S11, VSWR, Z_{in}, power to the load and radiated power per marker, and the worst case in the band. **Copy table**, **Download sweep (CSV)**, **Touchstone**.
- **Measurement**: draws a VNA file (.s1p / .s2p) over the simulation.
- **Apply to board** writes the values (and part numbers) to the footprints in one undo step.

> **to load** is the power that reaches the antenna port; **radiated** also includes the antenna's radiation efficiency. Use the band worst case below the table to compare variants.

5.2 Antenna report

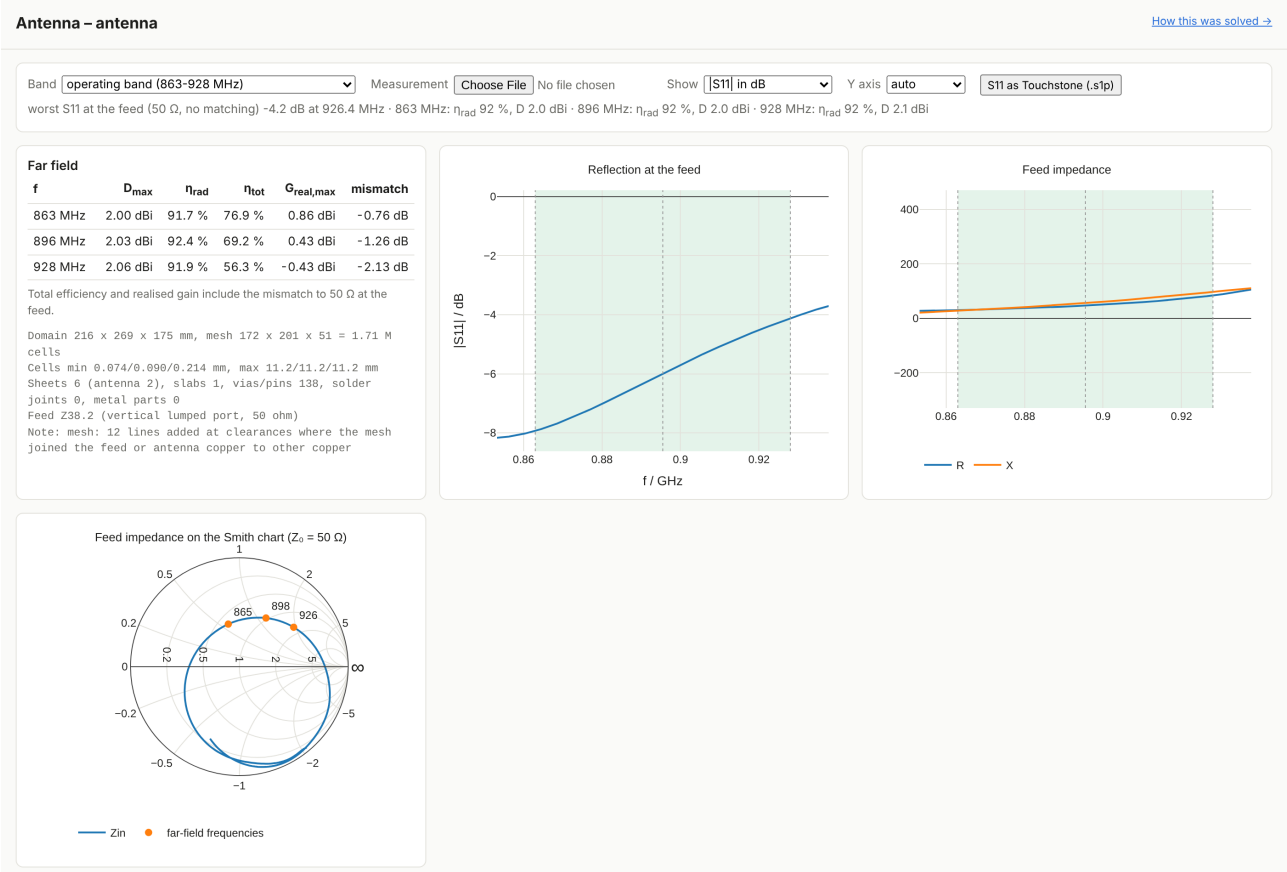


Figure 5: Antenna report: far-field table, reflection at the feed, feed impedance and Smith chart.

- **Far field** per frequency: D_{max}, radiation efficiency η_{rad}, total efficiency η_{tot} (includes the mismatch to 50 Ω), peak realised gain and mismatch.
- **Reflection at the feed**: |S11| in dB or linear, VSWR, return loss or mismatch loss. A measured .s1p can be drawn over it.
- **Feed impedance** (R, X) and the Smith chart with the far-field frequencies.
- The band line shows the worst S11 and η_{rad} / D inside the band.

These values belong to the bare antenna at the feed port, without the matching network. The tuning viewer shows the matched result.

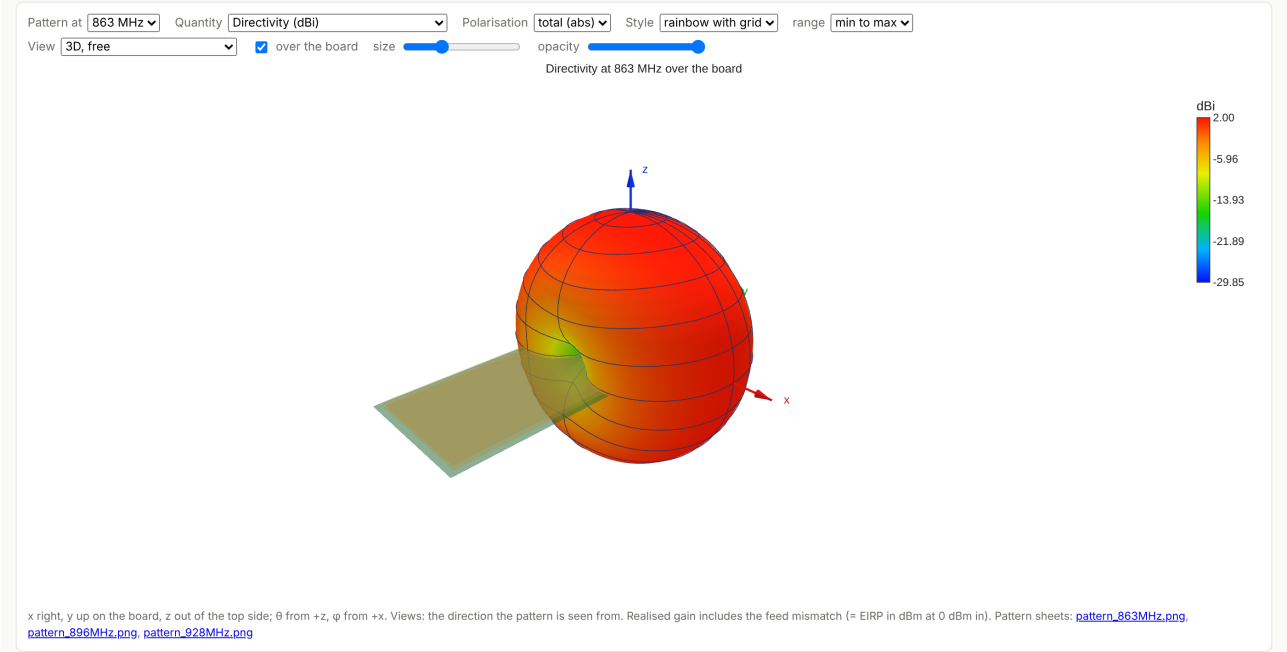


Figure 6: Radiation pattern over the board (directivity, rainbow scale with 15° grid).

- Axes: x right, y up on the board, z out of the top side. θ from $+z$, φ from $+x$.
- **Quantity**: directivity or realised gain. Realised gain includes the mismatch; at 0 dBm into the feed it equals the EIRP in dBm.
- **Polarisation**: total, E_θ or E_φ. **View**: free 3D, six standard views, or one.
- **over the board**: the pattern drawn around the antenna with board and casing.

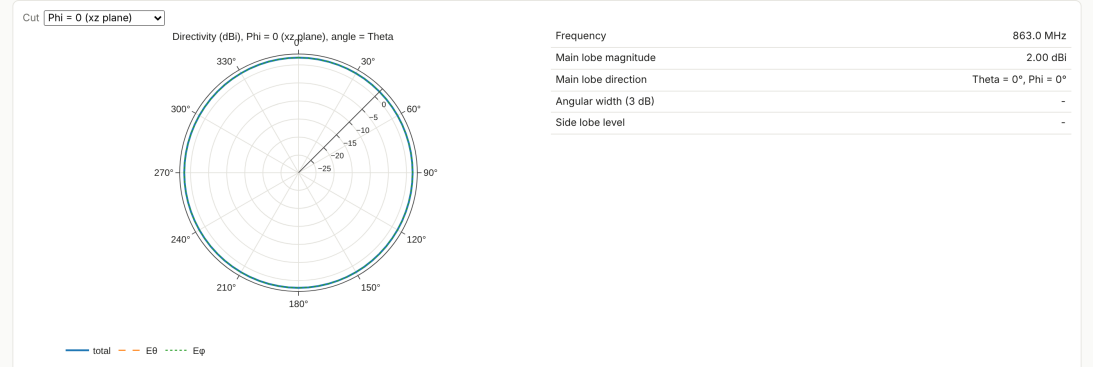


Figure 7: Cut through the pattern with main lobe, 3 dB width and side lobe level.

Each antenna run also writes `pattern_<f>MHz.png`: six views and three cuts on one sheet, for reports.

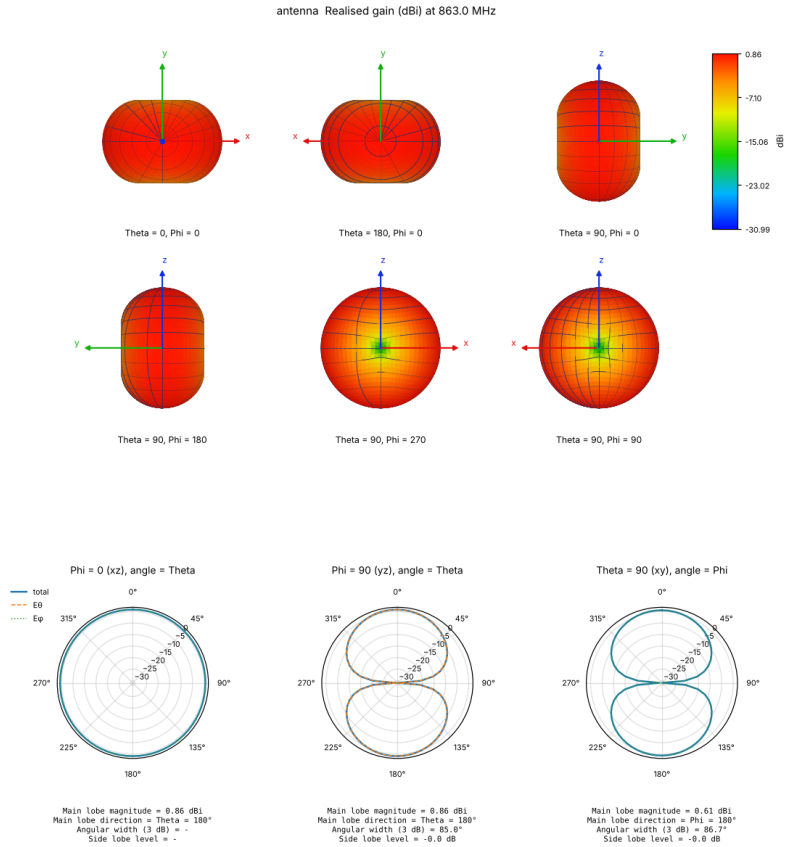


Figure 8: Pattern sheet `pattern_863MHz.png` (realised gain).

5.3 Mesh inspector

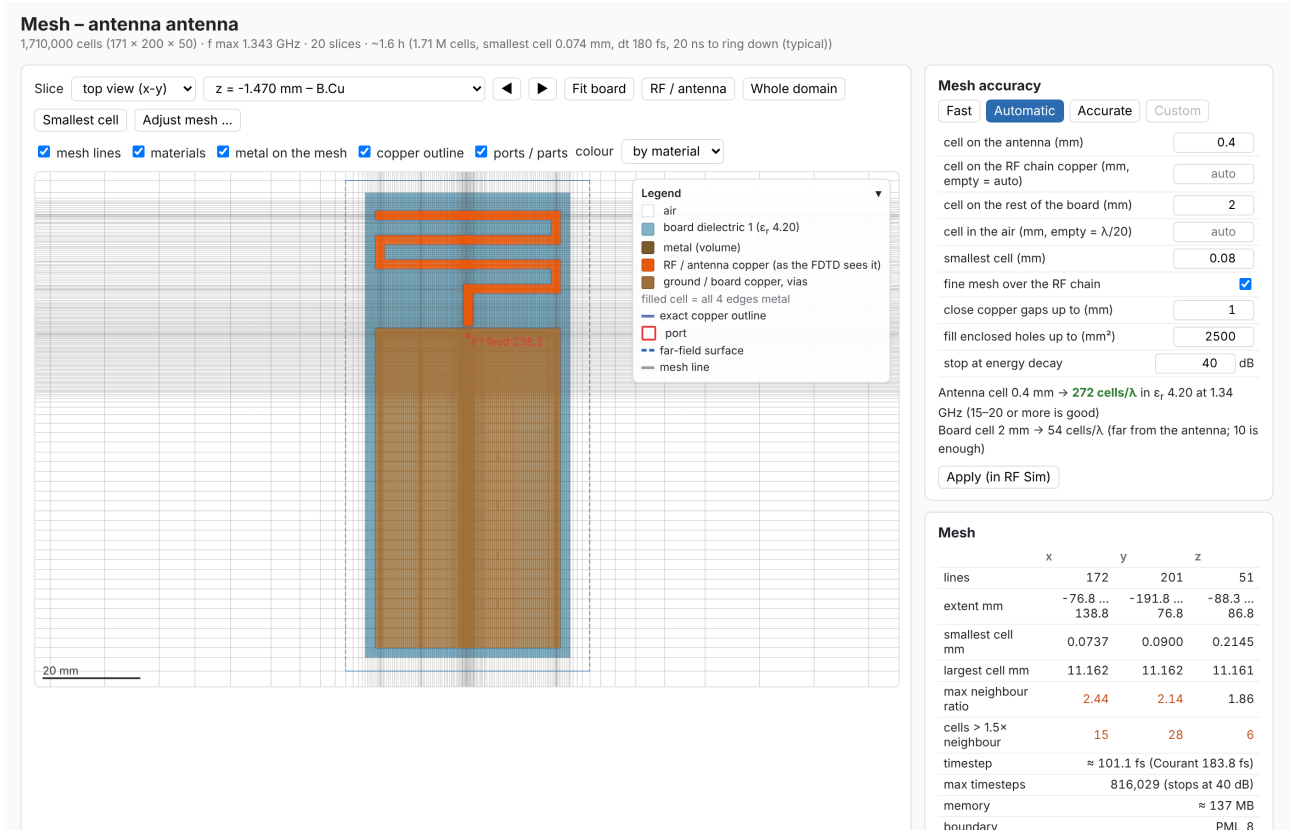


Figure 9: Mesh inspector: the model on its FDTD mesh, with the Mesh accuracy card.

Inspect Mesh (Setup page) builds the model exactly as the next simulation would. No simulation runs.

- **Slices**: top and side views through every copper plane, dielectric, port and the smallest cells.
- **Metal on the mesh**: the copper as the solver sees it. A trace that falls apart or a clearance that the mesh closes shows up here. *Copper outline* draws the exact copper on top.
- **Mesh accuracy**: change the cell sizes on the page. While you type it shows the cells per wavelength and a runtime factor. **Apply and rebuild mesh** saves and rebuilds.
- **Mesh** table and **Checks**: cells, smallest cell, timestep, memory, and the checks that also run before every simulation.

Good values: 15–20 or more cells per wavelength in the dielectric, and three or more cells across the narrowest RF trace.

5.4 Wave view

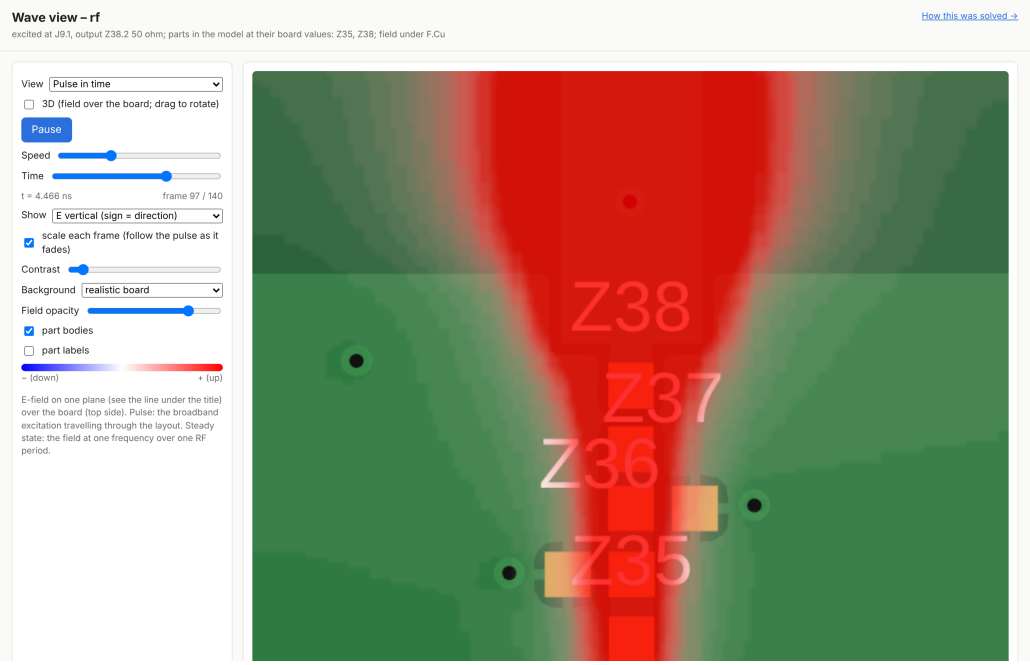
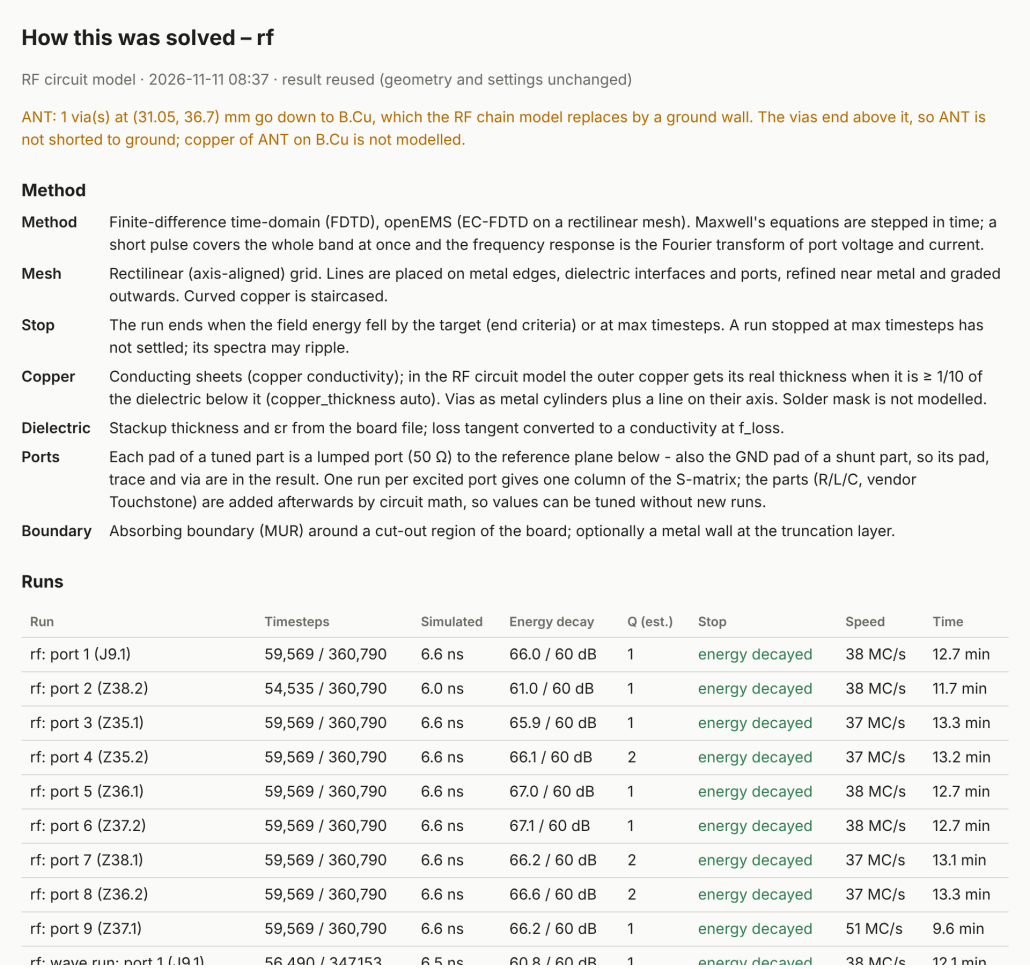


Figure 10: Wave view of the RF chain: the pulse travels from J9 through Z35 and Z38.

With **Wave view: record fields** each setup gets one extra run with all parts at their board values. The page animates the pulse in time or the steady state at one frequency, in 2D or 3D over the board. The *whole-board* view joins the RF chain and the antenna with one clock.

5.5 How this was solved



How this was solved – rf

RF circuit model · 2026-11-11 08:37 · result reused (geometry and settings unchanged)

ANT: 1 via(s) at (31.05, 36.7) mm go down to B.Cu, which the RF chain model replaces by a ground wall. The vias end above it, so ANT is not shorted to ground; copper of ANT on B.Cu is not modelled.

Method

Method	Finite-difference time-domain (FDTD), openEMS (EC-FDTD on a rectilinear mesh). Maxwell's equations are stepped in time; a short pulse covers the whole band at once and the frequency response is the Fourier transform of port voltage and current.
Mesh	Rectilinear (axis-aligned) grid. Lines are placed on metal edges, dielectric interfaces and ports, refined near metal and graded outwards. Curved copper is staircased.
Stop	The run ends when the field energy fell by the target (end criteria) or at max timesteps. A run stopped at max timesteps has not settled; its spectra may ripple.
Copper	Conducting sheets (copper conductivity); in the RF circuit model the outer copper gets its real thickness when it is ≥ 1/10 of the dielectric below it (copper_thickness auto). Vias as metal cylinders plus a line on their axis. Solder mask is not modelled.
Dielectric	Stackup thickness and εr from the board file; loss tangent converted to a conductivity at f_loss.
Ports	Each pad of a tuned part is a lumped port (50 Ω) to the reference plane below - also the GND pad of a shunt part, so its pad, trace and via are in the result. One run per excited port gives one column of the S-matrix; the parts (R/L/C, vendor Touchstone) are added afterwards by circuit math, so values can be tuned without new runs.
Boundary	Absorbing boundary (MUR) around a cut-out region of the board; optionally a metal wall at the truncation layer.

Runs

Run	Timesteps	Simulated	Energy decay	Q (est.)	Stop	Speed	Time
rf: port 1 (J9.1)	59,569 / 360,790	6.6 ns	66.0 / 60 dB	1	energy decayed	38 MC/s	12.7 min
rf: port 2 (Z38.2)	54,535 / 360,790	6.0 ns	61.0 / 60 dB	1	energy decayed	38 MC/s	11.7 min
rf: port 3 (Z35.1)	59,569 / 360,790	6.6 ns	65.9 / 60 dB	1	energy decayed	37 MC/s	13.3 min
rf: port 4 (Z35.2)	59,569 / 360,790	6.6 ns	66.1 / 60 dB	2	energy decayed	37 MC/s	13.2 min
rf: port 5 (Z36.1)	59,569 / 360,790	6.6 ns	67.0 / 60 dB	1	energy decayed	38 MC/s	12.7 min
rf: port 6 (Z37.2)	59,569 / 360,790	6.6 ns	67.1 / 60 dB	1	energy decayed	38 MC/s	12.7 min
rf: port 7 (Z38.1)	59,569 / 360,790	6.6 ns	66.2 / 60 dB	2	energy decayed	37 MC/s	13.1 min
rf: port 8 (Z36.2)	59,569 / 360,790	6.6 ns	66.6 / 60 dB	2	energy decayed	37 MC/s	13.3 min
rf: port 9 (Z37.1)	59,569 / 360,790	6.6 ns	66.2 / 60 dB	1	energy decayed	51 MC/s	9.6 min
rf: wave run: port 1 (J9.1)	56,490 / 347153	6.5 ns	60.8 / 60 dB	1	energy decayed	38 MC/s	12.1 min

Figure 11: How this was solved: warnings, method, and every run with its energy decay.

Every result has this page. It lists the warnings, the method and its simplifications, each run with its timesteps, energy decay and stop reason, the mesh statistics, every setting with its default, and the plausibility checks (reciprocity, passivity, energy balance, copper dropped by the solver).

A run that stopped at the time limit instead of the energy criterion has not settled. The page then gives the loaded Q and the time that would be needed.

5.6 Mesh convergence check

Is the mesh fine enough? On **Simulate** choose **Run: Mesh convergence check** (*Fast* + *Automatic* for a quick look, + *Accurate* for the full check). Results → **Show: Mesh convergence** lists the key values on each mesh and the change between the two finest meshes:

- antenna, per band: S11 minimum and its frequency, resonance ($X = 0$), R and X at the band centre, radiation efficiency, directivity;
- RF chain: |S11| and |S21| at the markers.

converged (✓) means the two finest meshes agree within the tolerances (frequency 1 %, S-parameters 0.5 dB, impedance 5 %, efficiency 3 points, directivity 0.3 dB). Otherwise use *Accurate* or a finer cell on the antenna.

6 Operating band

One band for all setups of a board, so the antenna and the RF chain use the same frequencies.

Sweep	0.6× the lowest to 1.5× the highest band centre, at least four points across the narrowest band (201–1001 points).
Far field, markers	band edges and centre of each band.
Part Q	the default Q (30 for L, 150 for C) refers to the band centre.
Presets	SRD 868, ISM 915, 868 + 915, ISM 433, ISM 2.4 GHz, 868 MHz + 2.4 GHz, LTE-M / NB-IoT, GNSS L1, Wi-Fi 5 GHz – or your own ranges.

Set it on the Setup page before **Create Setup**, or later with **Apply Band to All Setups**.

7 Good modelling practice

- **Feed the antenna with a short port.** A lumped port is a piece of wire: about 1 nH per mm. Feed at the connector pad with the ground plane below, or on the antenna trace right after the matching network. RF Sim warns when the port spans more than 1.5 mm.
- **Select only the parts you tune.** Fewer ports mean fewer runs. Fixed parts inside the region are modelled as lumped parts.
- **Start with Fast, finish with Automatic or Accurate.** Run the convergence check once per new design.
- **Measurement fixtures.** A coax cable on a small board is part of the antenna. Compare with measurements made with ferrite chokes on the cable.
- **Ground.** A ground net with another name (`AGND`, `VSS`, ...) is recognised. On a 2-layer board the bottom layer is the reference of the RF chain; vias of other nets that go down to it are cut there (warning on the solver page).
- **Look at the warnings.** The Setup page, the log and *How this was solved* name what to check, e.g. a region edge that cuts an RF trace or a mesh that joins the feed to other copper.

8 Files and folders

`<project>/rfsim/rf.json`, `antenna.json`	setups (JSON, editable).
`<project>/rfsim/enclosure.json`	casing parts.
`<project>/rfsim/results/<setup>/`	results: `tune.html`, `antenna_report.html`, `mesh.html`, `wave.html`, `solver_info.html`, Touchstone files, `pattern_<f>MHz.png`.
`%LOCALAPPDATA%\kicad-rfsim`	openEMS, model cache, board snapshots (removed after a day).

The result pages are self-contained HTML. They open in any browser; **Save Page as Single File** makes a copy that works without the shared `plotly.min.js`.

9 Troubleshooting

Symptom	What to do
The RF Sim button is missing	Enable the KiCad API (**Preferences** → **Plugins**) and restart KiCad.
"openEMS not found"	**Solver Settings** → **Browse** or **Search Again**; or let RF Sim download it.
Create Setup refuses	Choose the operating band first.
No antenna detected	Use **Define antenna**: pick the net and the feed pad.
S21 of 0 Ω parts far below 0 dB	Inspect the mesh: a short or an open on the mesh, or a via of the RF net to the reference layer (see the solver page warning).
Fast and Automatic disagree strongly	Long feed port or RF copper beside the feed; feed at the connector or on the antenna trace. Run the convergence check.
A run hit the time limit	Loaded Q too high for the limit; raise `max_time_ns` (Custom) as the solver page suggests.
"numerically unstable"	Use Automatic, or a larger smallest cell.
Simulation is slow	Use Fast for a first look; select fewer parts; close other simulations.

10 Accuracy and limits

RF Sim has been checked against closed-form references (microstrip and coplanar lines, a $\lambda/4$ stub, single parts, a dipole, a patch) and against measured antennas of a public reference kit (433 MHz to 2.4 GHz). Typical agreement: line impedance

within 3 %, resonance frequencies within 1–4 %, total efficiency mostly within the chamber uncertainty of ± 3 dB. Details: `docs/VALIDATION.md`.

Not modelled:

- solder mask (lowers a printed antenna's resonance by about 1–2 %);
- cables and fixtures;
- part self-resonance and frequency-dependent Q (unless vendor Touchstone files are used);
- material dispersion and copper roughness;
- curved copper is staircased on the mesh; rotations of antenna boards in 90° steps only.